

QueryBook — Drawings, Reference Numerals & Brief Description of the Drawings

Scope: the complete, current QueryBook figure set — FIG. 1–FIG. 34 — as one unified set: the integral language & speech subsystem (LEL; FIG. 1–8, 31–32), the QueryBook platform (FIG. 9–30), and the Language & Identity Linking (LIL) voiceprint & identity subsystem (FIG. 33–34). **Companion** to the consolidated *QueryBook Provisional Patent Application*; the figures, reference numerals, and brief descriptions here are identical to those in that application. **Document date:** October 1, 2026. Draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman before filing.

Brief Description of the Drawings

FIG. 1 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 2 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 3 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 4 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 5 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 6 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 7 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

FIG. 8 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

Brief Description of the QueryBook Platform Drawings (FIG. 9–FIG. 30)

FIG. 9 is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

FIG. 10 is a diagram of the fact Unit — atom, envelope, and content-derived identity.

FIG. 11 is a diagram of the fact-certification authority hierarchy.

FIG. 12 is a diagram of the uFCS substrate — three-store architecture.

FIG. 13 is a diagram of the authored tour and exhibition sequencing.

FIG. 14 is a diagram of the fQL — joint structure and trust execution.

FIG. 15 is a diagram of the verifying machine-generated text against certified records.

FIG. 16 is a diagram of the federation — cross-boundary re-validation.

FIG. 17 is a diagram of the fact-level erasure by external-key destruction.

FIG. 18 is a diagram of the universal security architecture.

FIG. 19 is a diagram of the universal honeypot saturation and data firehose.

FIG. 20 is a diagram of the deployment topologies over an invariant common layer.

FIG. 21 is a diagram of the adaptive response engine — directive fixed before generation.

FIG. 22 is a diagram of the user-owned agent system with governance and kill switch.

FIG. 23 is a diagram of the deterministic multi-language regeneration.

FIG. 24 is a diagram of the semantic compression, expansion, colorization and upscaling.

FIG. 25 is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

FIG. 26 is a diagram of the neuromorphic cognitive layer — attractor convergence.

FIG. 27 is a diagram of the fact Unit — confidence and evidence model.

FIG. 28 is a diagram of the uFCS packet structure — serialized Fact Unit record.

FIG. 29 is a diagram of the uFCS packet framing — content-addressed block storage.

FIG. 30 is a diagram of the fQL — query anatomy and worked example.

Reference Numerals

No.	Element	No.	Element
100	QueryBook LEL system	870	optional future dictionary-gated suggestor
110	Sub-Language Priming Layer (SLPL)	900	advanced multilingual pronunciation subsystem
120	Language Expression Layer (LEL)	910	pronunciation request (word/phrase, language)
130	Transition Gate	920	pronunciation preference-ladder selector
140	Phase Two semantic-communicative learning	930	bundled open-source phonemizer
150	Multilingual Delta Acquisition module	931	phonemizer IPA transcription
160	Speech Output Pipeline	932	phonemizer call cache
165	natural speech output	933	bounded call log / timeout guard
170	co-grounding interface	940	rule-seeded G2P tables (per language)
180	Federated Query Language (FQL)	941	approximate rule-seeded IPA (offline)
190	Universal Federated Connectivity System (UFCS)	950	syllable count + primary-stress derivation
195	federated data sources	960	explicit null result (not fabricated)
200	SLPL	970	phrase-level single-call transcription
210	acoustic phonetic processing	971	displayed transcription consistent with synthesized audio
211	signal-level feature embedding	980	self-contained application bundle
220	phonics / phonemic awareness	981	bundled open-source speech engine
221	phoneme-boundary segmenter	982	runtime-support libraries
230	communication-science foundations	983	bundled phoneme data
231	communicative-salience weight	984	engine-selection ladder (per OS / language)
240	prosody & rhythm processing	985	host OS built-in voice service
241	prosodic contour vector	986	culture-matched OS voice (when installed)
250	information-theoretic grounding	987	honest refusal with remedy (no fabricated audio)
251	information weighting	990	deterministic translator (through English)
260	neurolinguistic chunking	991	bundled open-source bilingual dictionary
261	chunk segments	992	coverage accounting + unknown-word passthrough (flagged)
270	licensed multimedia input	993	deterministic language detector
300	developmental learning sequence	995	permanent quality-control battery
310	Phase One (structural-acoustic)	996	IPA-produced check
311	sound discrimination	997	syllable-sanity check (± 1)
312	syllabic segmentation	998	valid-waveform synthesis check
313	word-as-acoustic-object	999	per-language QC report + overall pass score
314	semantics suppressed	1000	LIL voiceprint & identity subsystem
320	Transition Gate (one-way)	1001	consent gate (precondition)
321	phonemic-inventory completeness $\geq \tau_1$	1002	audio/video capture + VAD/segmentation
322	lexical stability $\geq \tau_2$	1003	acoustic feature extraction
323	co-occurrence density $\geq \tau_3$	1004	speaker embedding (voiceprint vector)
324	pattern saturation $< \epsilon$	1005	enrollment — canonical voiceprint
330	Phase Two (semantic-communicative)	1006	match / score (cosine · PLDA)
331	word-to-referent grounding	1007	QueryBook trust gate (VERIFIED/CONTRADICTED/UNKNOWN)
332	communicative discovery	1008	Fact-Unit identity graph
333	emergent grammar / syntax	1009	LEL cross-script name/entity resolution
400	Delta Acquisition Model	1010	provenance-tracked link Fact Unit
410	reused L1 foundation	1020	voice authentication request
420	L2 delta detector	1021	capture phrase + embed
430	targeted training of novel elements	1022	anti-spoofing (liveness/replay/synthetic)
440	multilingual semantic alignment	1023	deterministic risk-scoring policy
450	acquisition acceleration flywheel	1024	authentication decision gate
451	negative-transfer detection & retraining	1025	second-factor challenge
500	Speech Output Pipeline	1026	auth-event Fact Unit (provenance)
510	lexical selection	1100	UIAS integrity audit service
520	grapheme-to-phoneme conversion	1101	monitored system workflow (unmodified)
530	coarticulation modeling	1102	input-boundary position
540	prosody generation	1103	processing-layer position
550	register & affect calibration	1104	storage-layer position
560	self-monitoring feedback loop	1105	output-boundary / handoff position
570	neural vocoder rendering	1106	Connector + Ingestion agents (seal at observation)

No.	Element	No.	Element
580	natural speech waveform	1107	seal = Fact Unit (hash · id-hash · independent timestamp · source · agent)
600	co-grounded integration	1108	Similarity Detection Agent (semantic signatures)
610	natural-language input	1109	Fact-Unit audit ledger (UFCS, append-only, immutable)
620	LEL analysis	1110	contemporaneous cross-party anomaly flag
630	FQL operations	1111	Verification Agent (continuous re-verification)
640	UFCS execution	1112	content-not-stored / one-way identity (fact-level erasure)
650	federated sources	1120	UIAS signed Proof Package generation
660	results	1121	accusation / inquiry
670	render to NL / speech in input language	1122	Proof Package Agent
680	language-agnostic resolution	1123	sealed Fact Units + verified seals
700	Language Lab prototype embodiment	1124	cross-submission similarity log entry
710	corpus source	1125	chronological ledger extract (RPH)
720	Phase-One structural agent	1126	Arbitration Agent (factual determination; Prime-Directive gate)
730	structural analyzer	1127	assemble + sign under UIAS independent identity
731	grapheme inventory	1128	signed Proof Package
732	rule-seeded G2P	1129	factual-only determination (legal reserved to courts/regulators)
733	syllabifier	1200	PIL paralinguistic & pragmatic interpretation
734	lexical-stability metric	1201	ingested voice (consent-gated)
735	co-occurrence density metric	1202	transcription (words)
736	prosody proxy	1203	Paralinguistic Analyzer (measured)
740	Fact Unit store	1204	tone/loudness/pacing/emphasis + labeled arousal estimate
750	Transition-Gate metrics	1205	Pragmatic Framer (sourced)
751	one-way latch	1206	context · perspective · ergonomics · culture · social mores
760	phase agents on shared manager	1207	Interpretation Frame (labeled estimates + sourced frames)
761	Phase-1 agent (implemented)	1208	Fact Units on ingested content
762	Phase-2 agent — deterministic dictionary + store grounding (implemented)	1209	covenant: estimates not asserted; frames sourced; consent
763	Phase-3 agent (declines — spec-only)	1210	PIL human query interpretation
764	Phase-4 agent (declines — spec-only)	1211	human query (typed/spoken)
800	deterministic Phase-Two semantic grounding	1212	Interpretation Frame (para + pragmatic)
810	Phase-One vocabulary (learned words)	1213	intent disambiguation (context/perspective)
820	deterministic grounder (per word)	1214	ambiguity gate
830	bundled public-domain dictionary source	1215	ask one clarifying question / stop
831	dictionary meaning Fact Unit (means, trust 0.9)	1216	register & ergonomic mode selector
840	self-grounding store-lookup source	1217	culture/mores/ergonomics packs
841	store-link Fact Unit (grounded_by_fact)	1218	retrieve-gate-compose (verified Fact Units; cite or refuse)
850	Fact Unit store (provenance, domain language)	1219	answer in selected register/mode (facts unchanged)
860	excluded large-language-model meaning source		

Drawings

Formal patent drawings — black line art, forming **one QueryBook figure set**: FIG. 1–FIG. 8 and FIG. 31–FIG. 32 depict QueryBook’s integral developmental language and speech subsystem (the Language Expression Layer), FIG. 9–FIG. 30 depict the QueryBook platform in which that subsystem is co-grounded (see FIG. 6), FIG. 33–FIG. 34 depict the Language & Identity Linking (LIL) voiceprint and multimodal identity subsystem, FIG. 35–FIG. 36 depict the Universal Integrity Audit Service (UIAS) that QueryBook delivers on these primitives, and FIG. 37–FIG. 38 depict the Paralinguistic & Pragmatic Interpretation Layer (PIL). Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

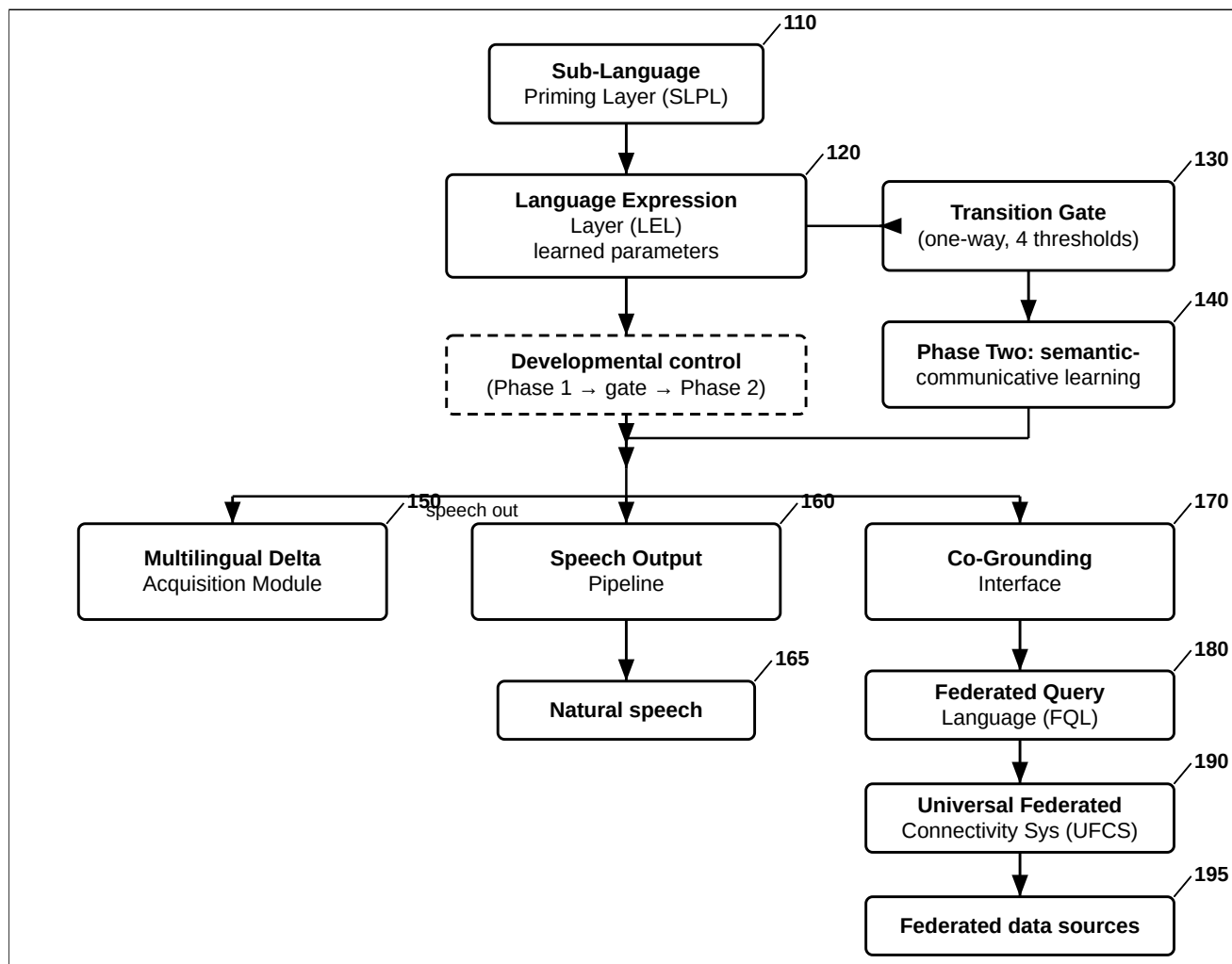


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

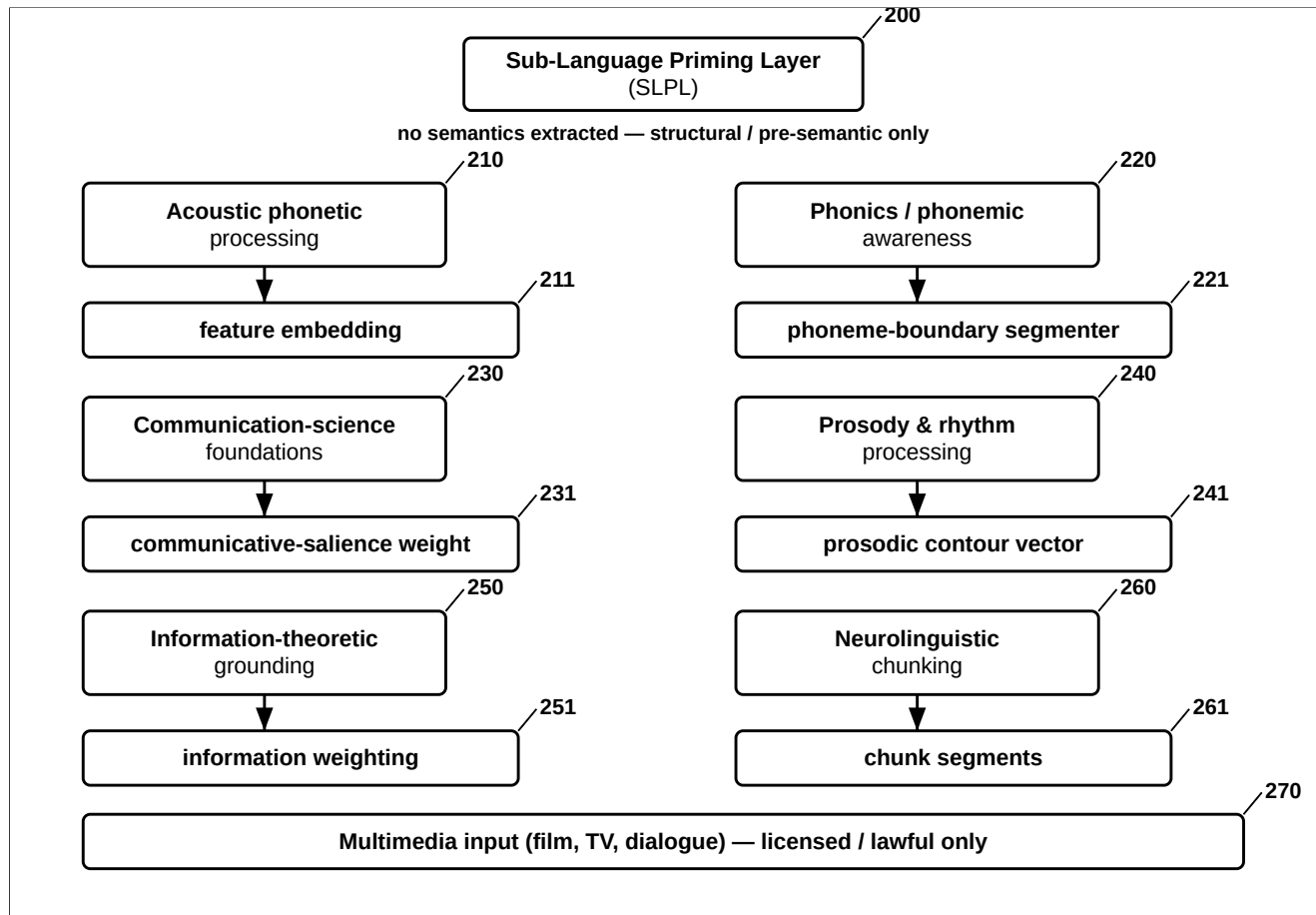


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

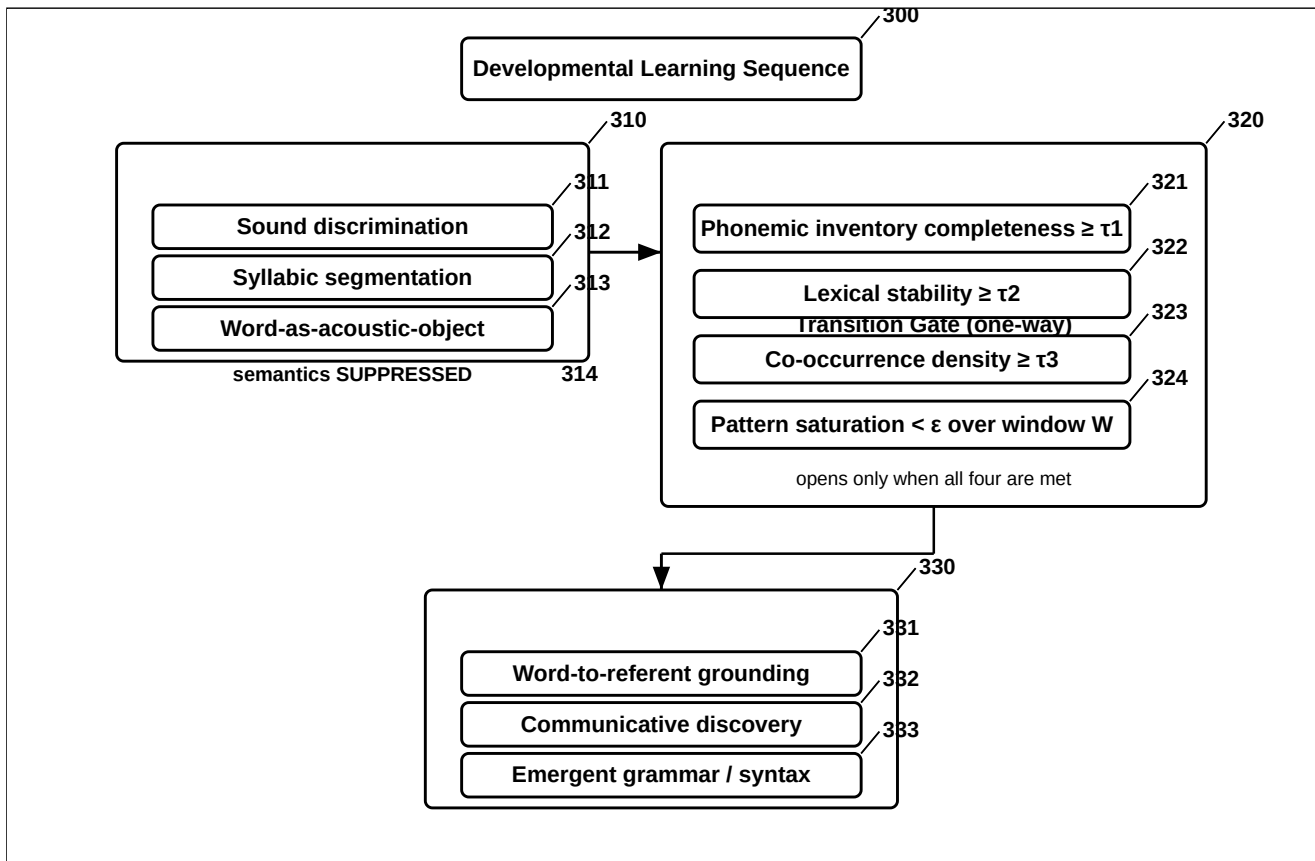


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

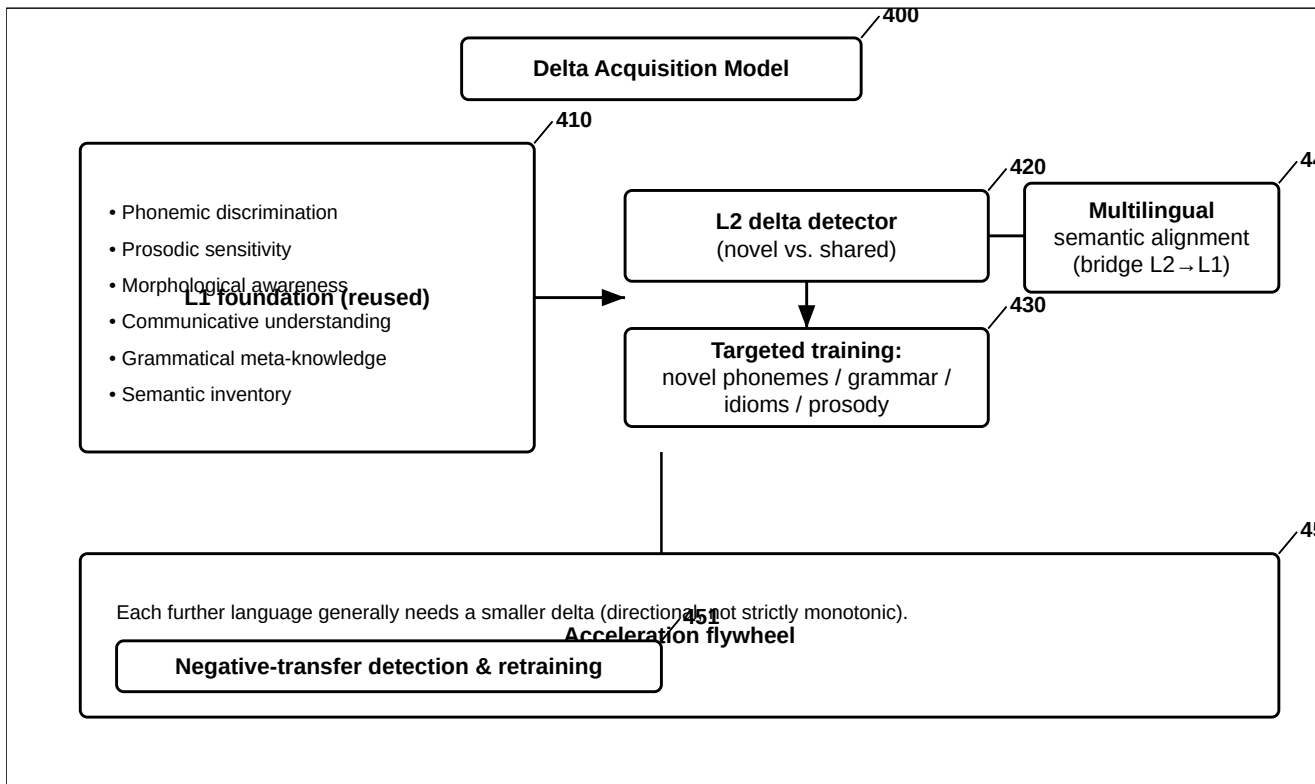


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

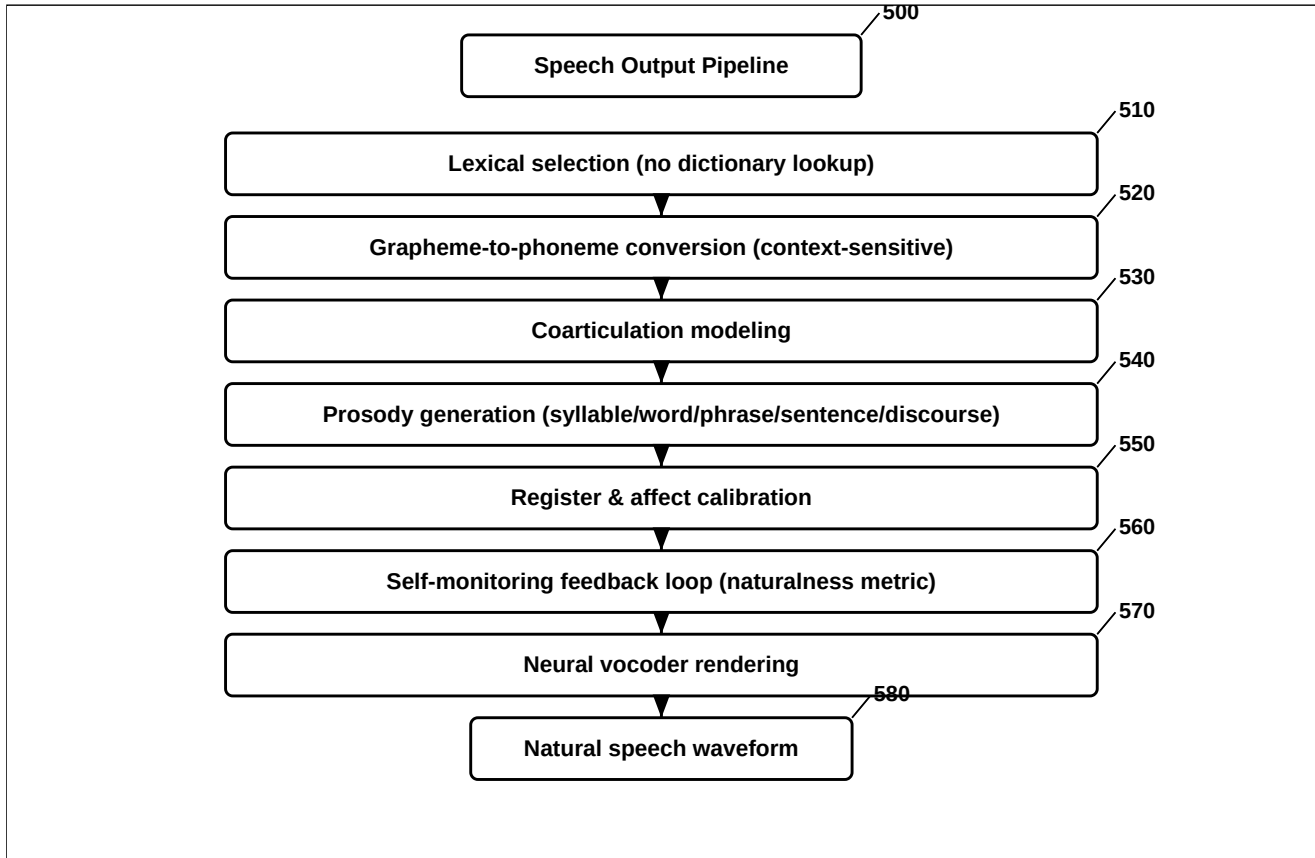


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

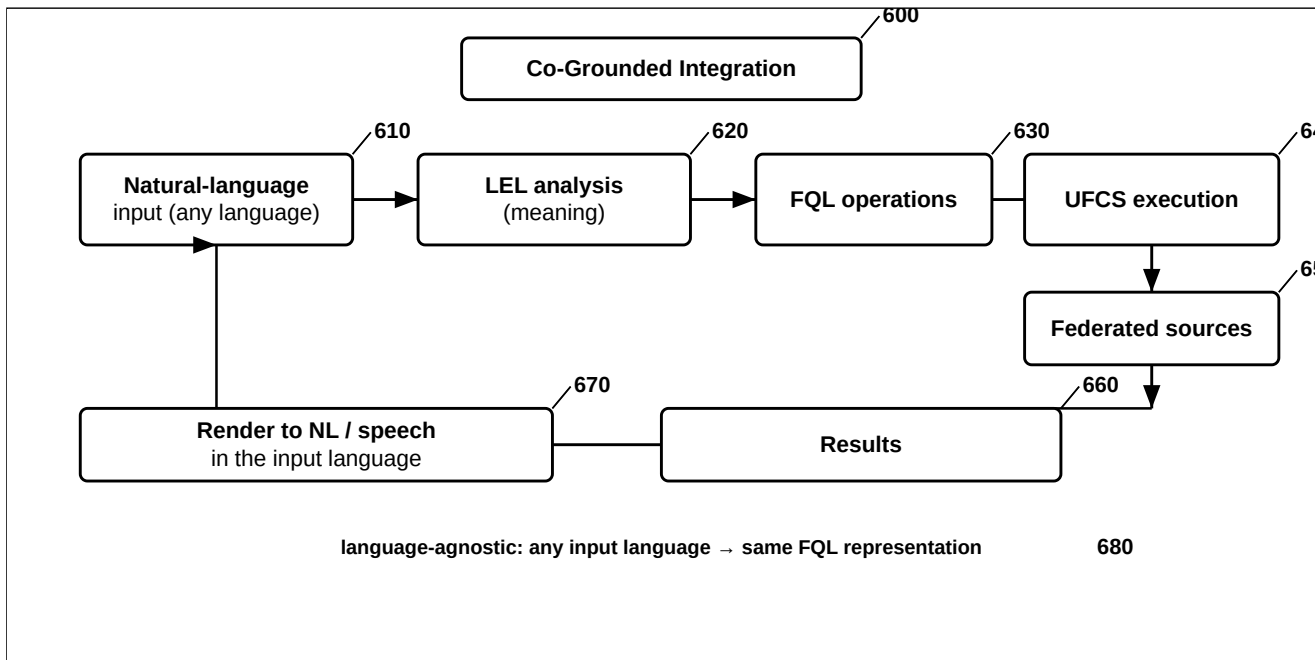


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

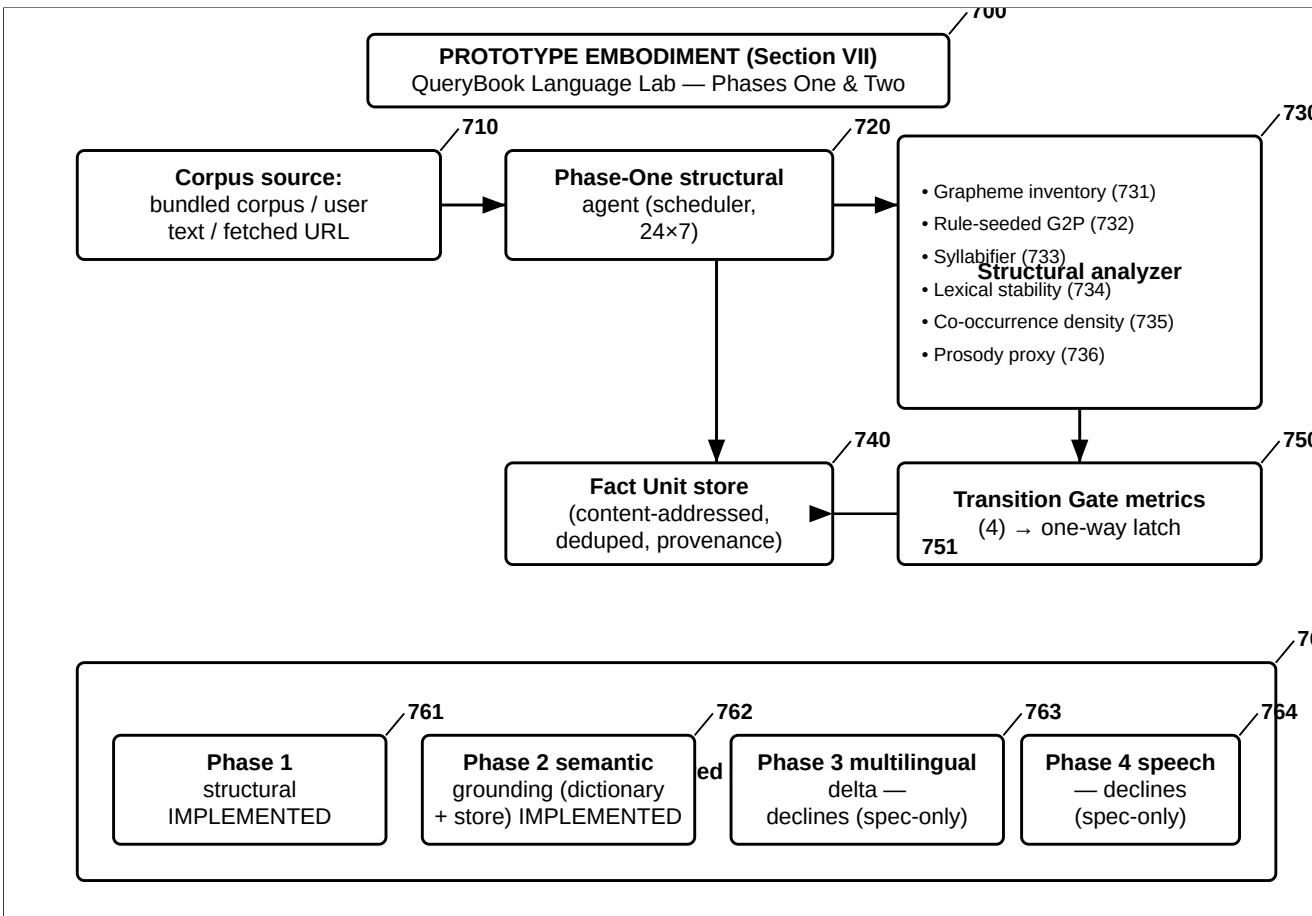


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

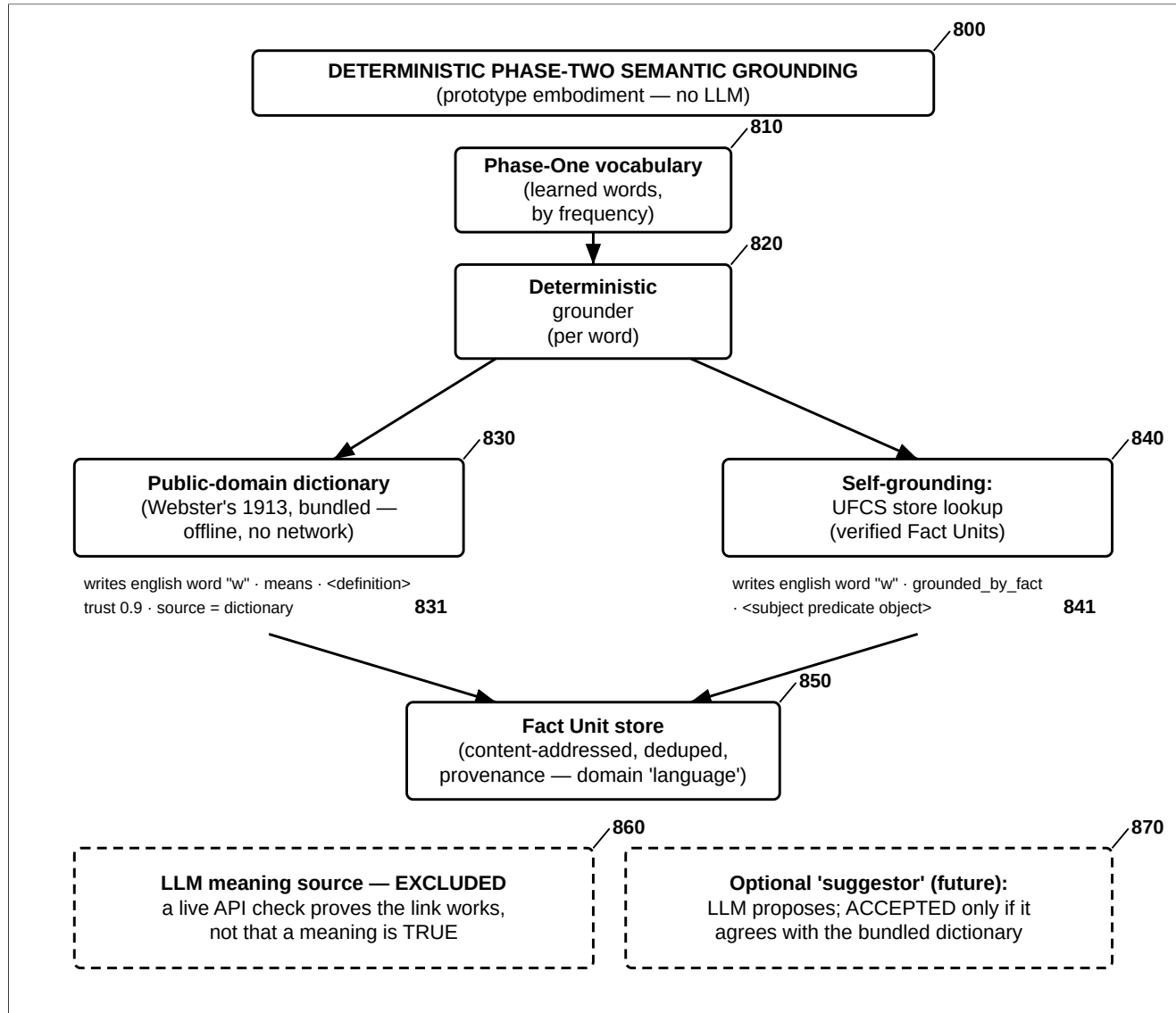


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

QueryBook Platform Drawings (FIG. 9–FIG. 30) — the system in which the language subsystem is integral

FIG. 9 through FIG. 30 depict the QueryBook deterministic, provenance-tracked federated knowledge and query platform of which the developmental language and speech subsystem (the Language Expression Layer, FIG. 1–FIG. 8 and FIG. 31–FIG. 32) is an integral, co-grounded part (see FIG. 6). Together with the language-subsystem figures these form **one QueryBook figure set**. Draft — the language-subsystem figures use a 100–999 numeral series and the platform figures a 100–2200 series; counsel/draftsperson to consolidate the two into a single QueryBook numeral series and finalize to 37 CFR 1.84 before filing.

FIG. 9

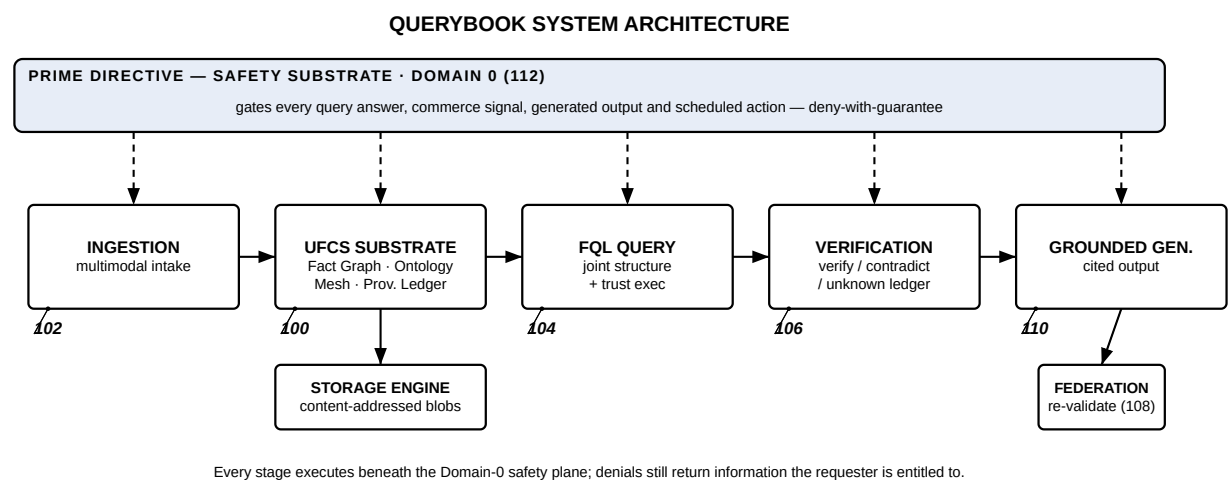


FIG. 9

FIG. 9 — QueryBook system architecture (Prime Directive as Domain 0)

FIG. 10

FACT UNIT — ATOM, ENVELOPE, AND CONTENT-DERIVED IDENTITY

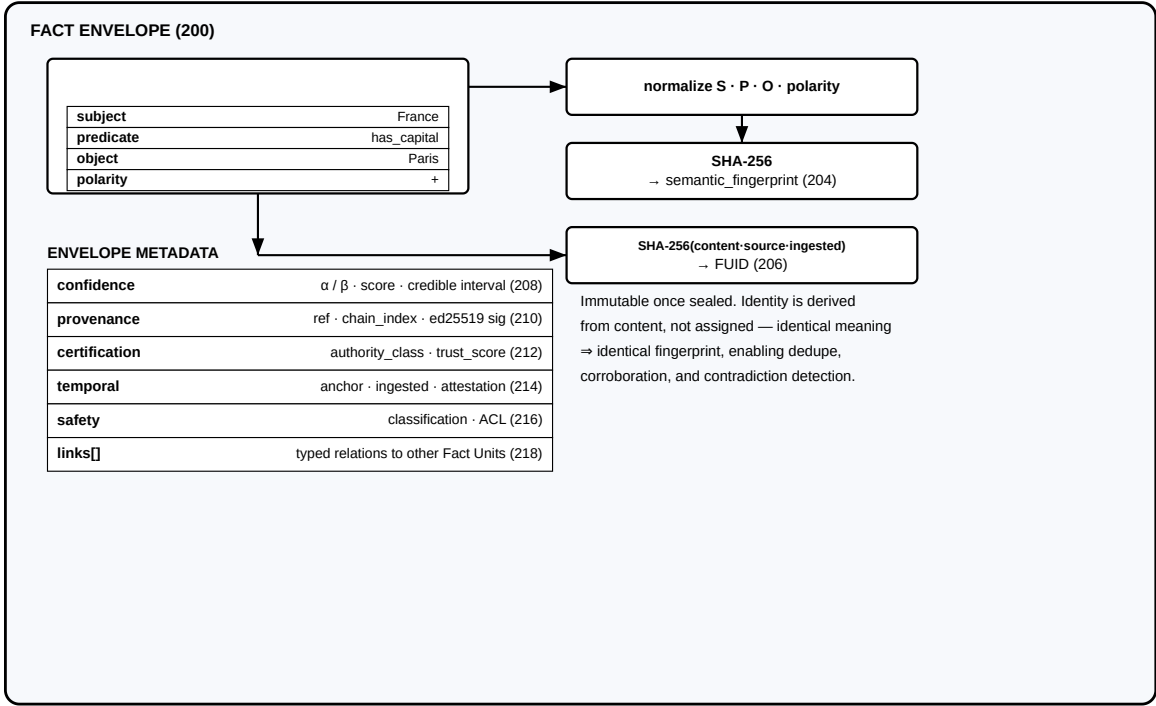


FIG. 10

FIG. 10 — Fact Unit — atom, envelope, and content-derived identity

FIG. 11

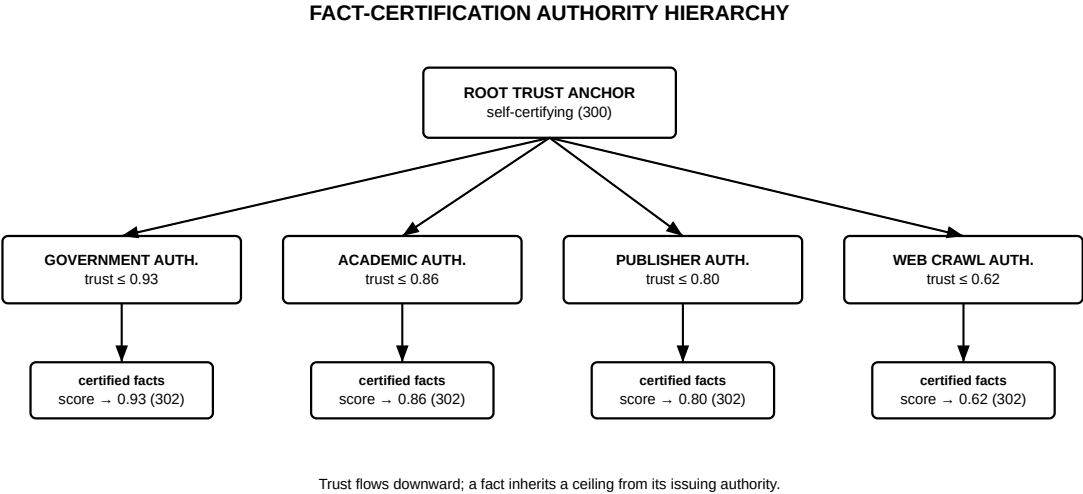


FIG. 11

FIG. 11 — Fact-certification authority hierarchy

FIG. 12

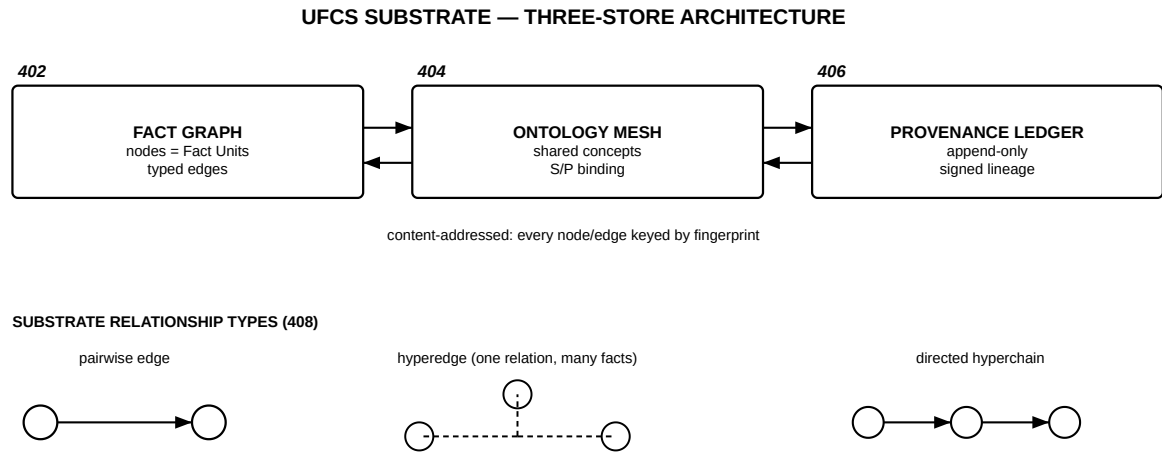
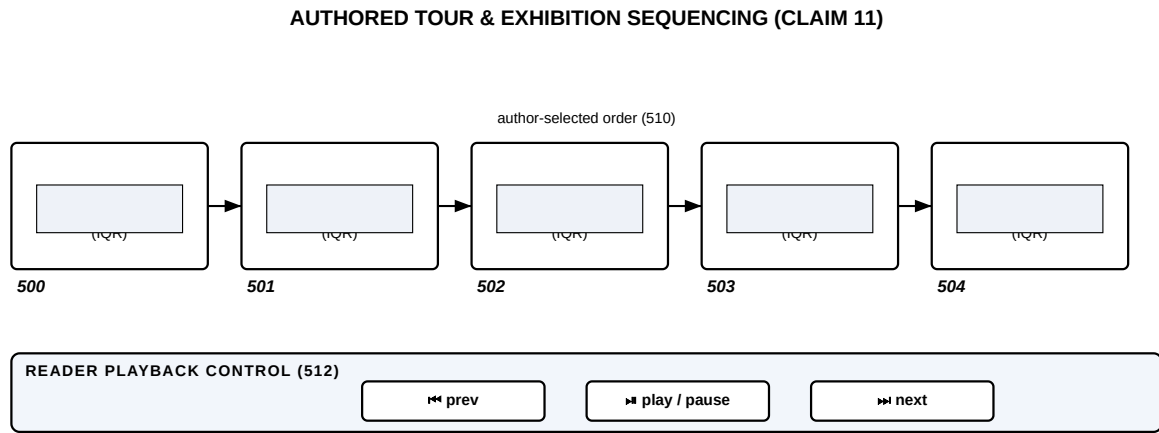


FIG. 12

FIG. 12 — UFCS substrate — three-store architecture

FIG. 13



Author fixes the sequence; the reader controls pace and direction along the ordered waypoints.

FIG. 13

FIG. 13 — Authored tour and exhibition sequencing

FIG. 14

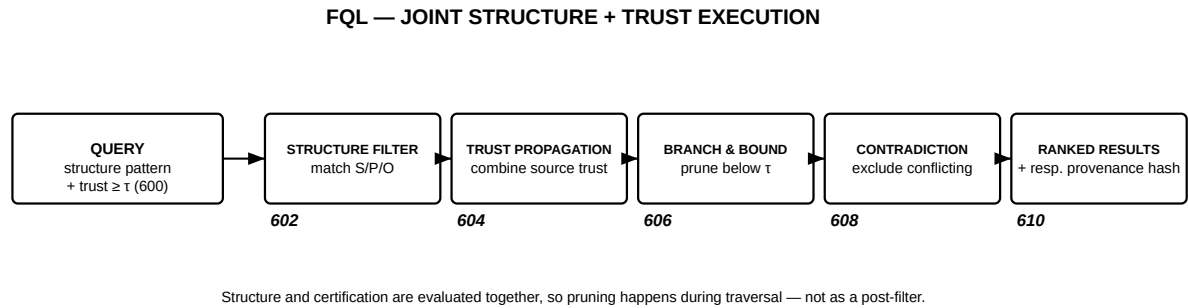


FIG. 14

FIG. 14 — FQL — joint structure and trust execution

FIG. 15

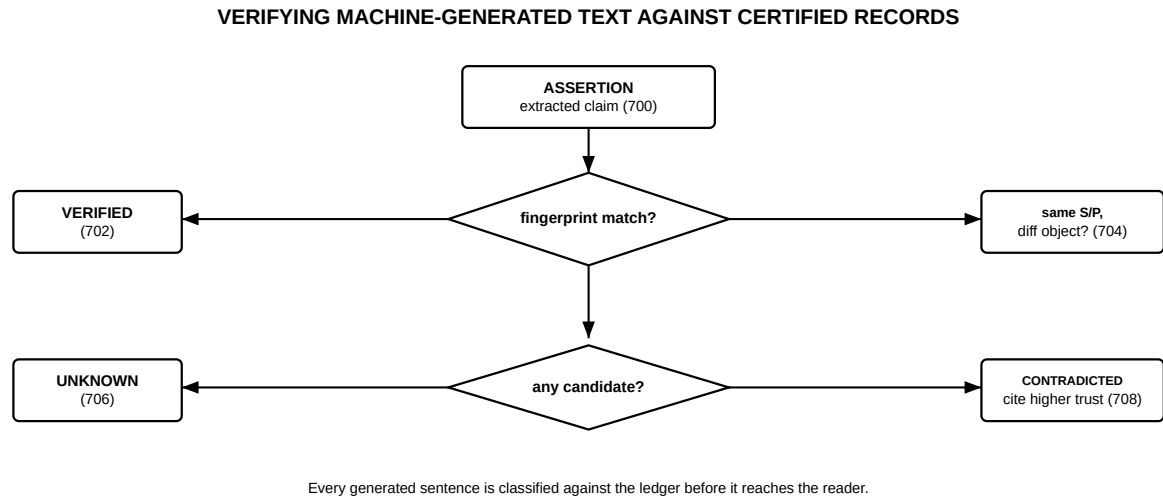


FIG. 15

FIG. 15 — Verifying machine-generated text against certified records

FIG. 16

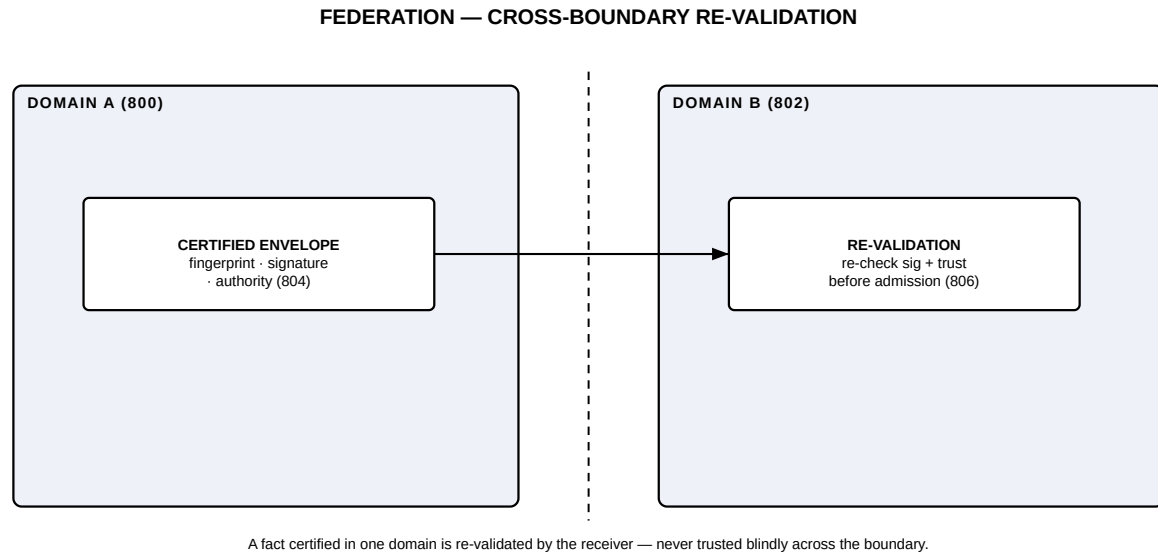
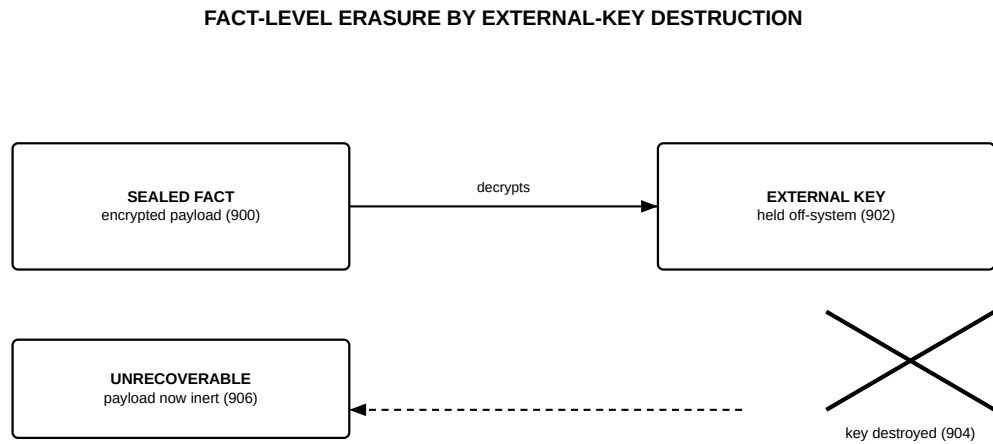


FIG. 16

FIG. 16 — Federation — cross-boundary re-validation

FIG. 17



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 17

FIG. 17 — Fact-level erasure by external-key destruction

FIG. 18

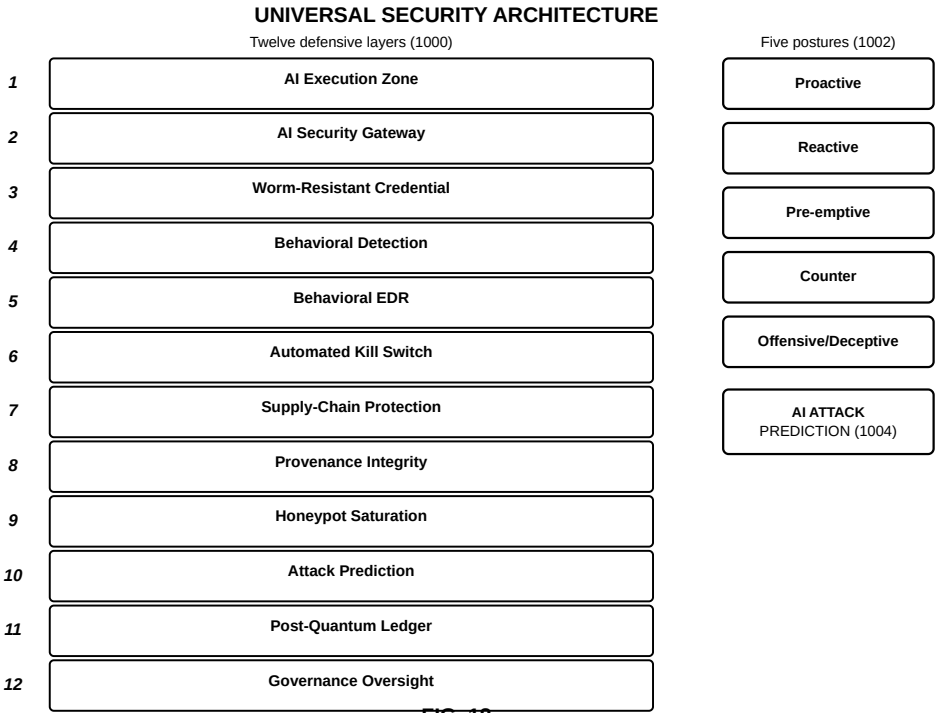


FIG. 18 — Universal security architecture

FIG. 19

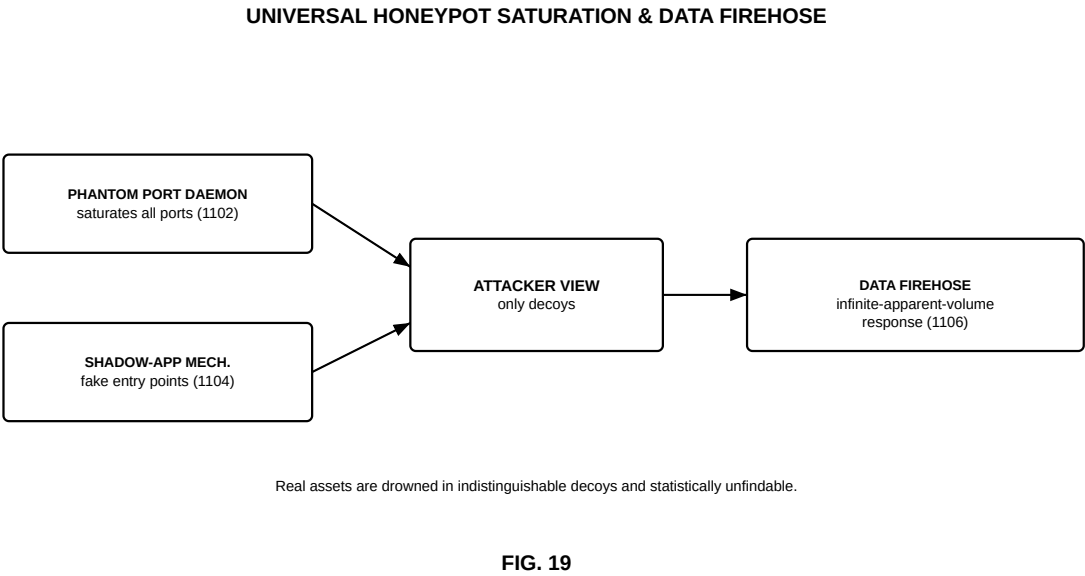


FIG. 19 — Universal honeypot saturation and data firehose

FIG. 20

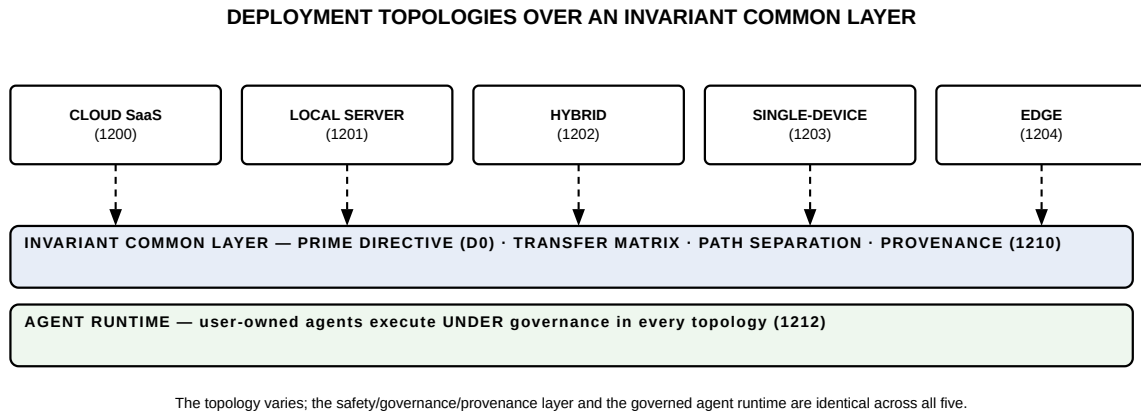


FIG. 20

FIG. 20 — Deployment topologies over an invariant common layer

FIG. 21

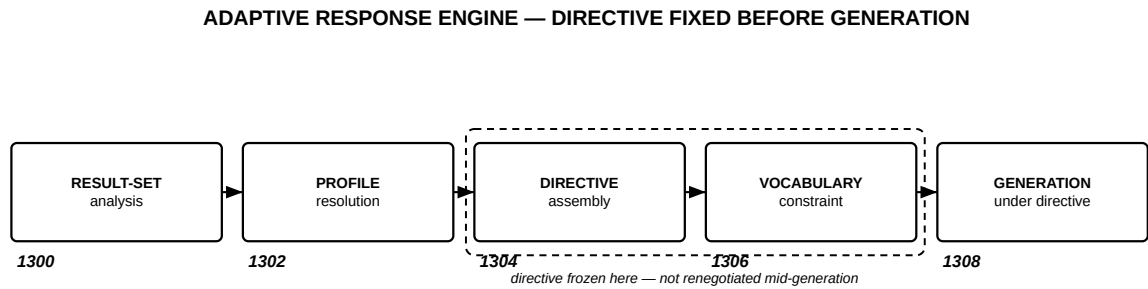
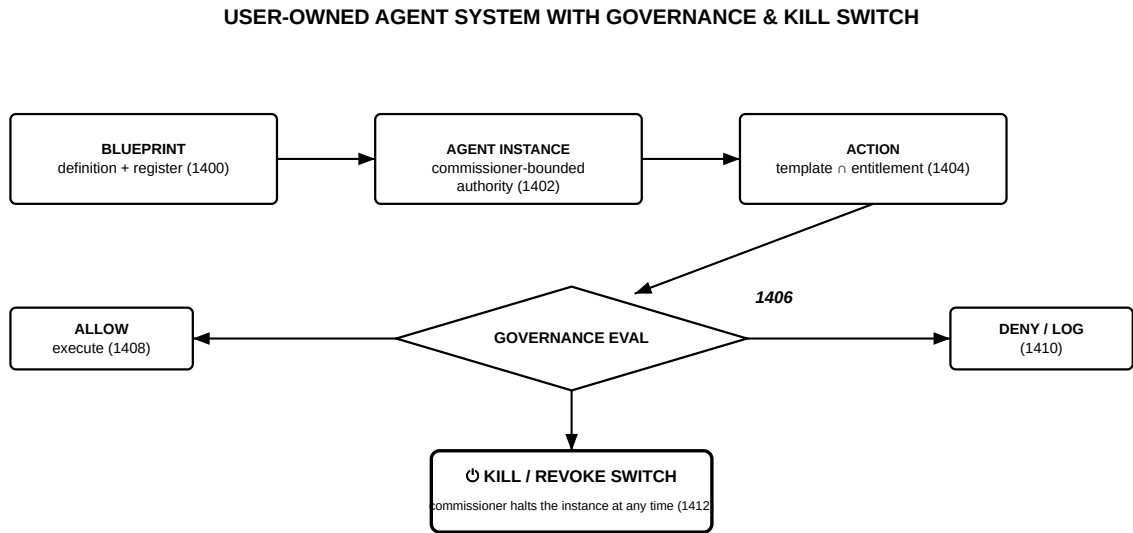


FIG. 21

FIG. 21 — Adaptive response engine — directive fixed before generation

FIG. 22

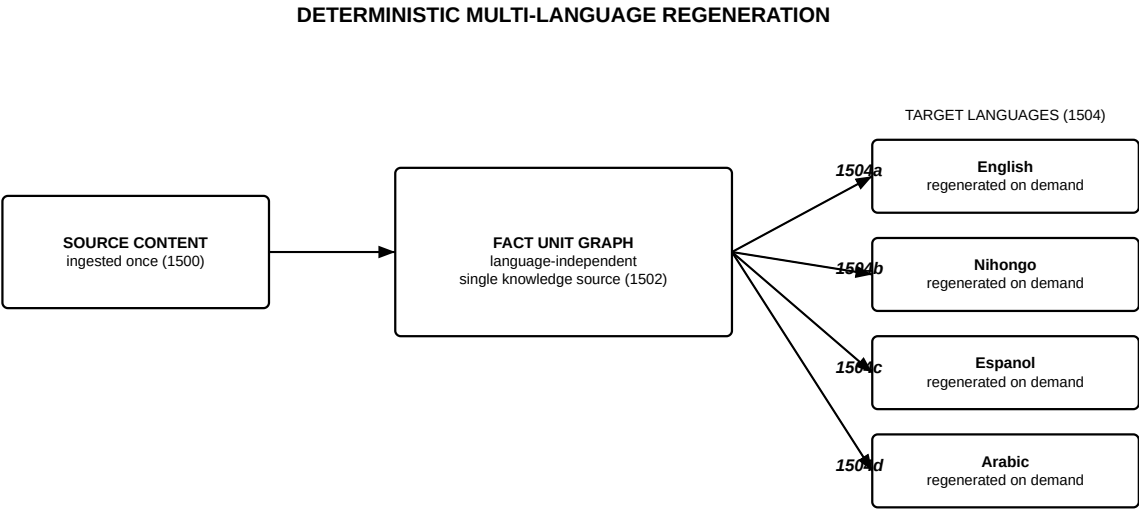


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 22

FIG. 22 — User-owned agent system with governance and kill switch

FIG. 23



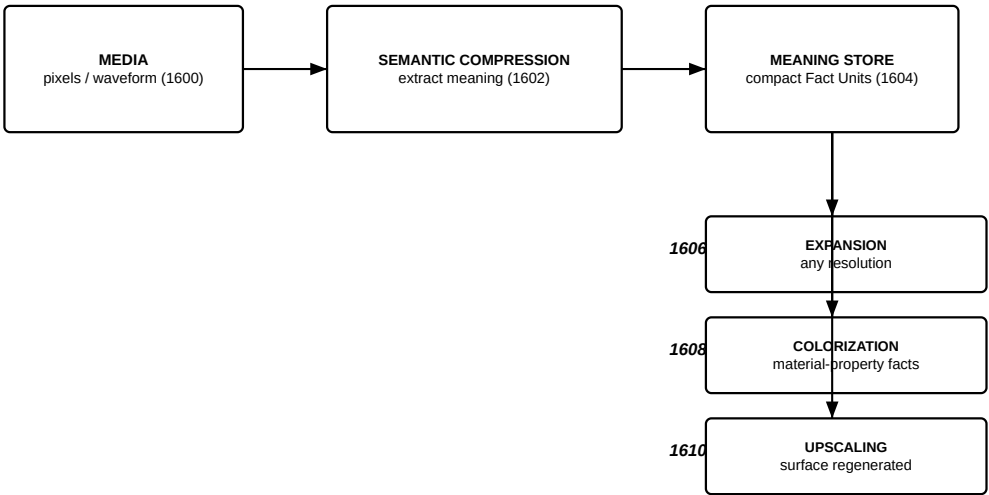
One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 23

FIG. 23 — Deterministic multi-language regeneration

FIG. 24

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING



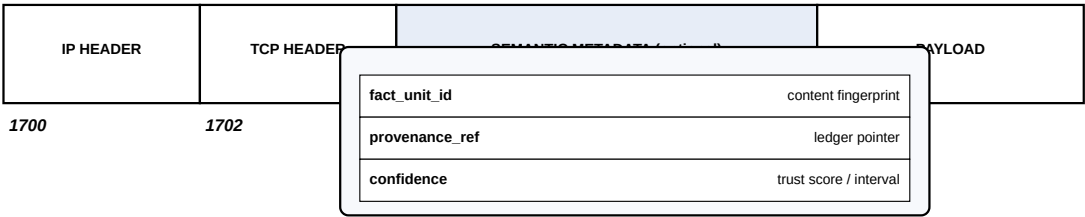
Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 24

FIG. 24 — Semantic compression, expansion, colorization and upscaling

FIG. 25

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 25

FIG. 25 — QueryBook-enhanced TCP/IP hybrid protocol packet

FIG. 26

NEUROMORPHIC COGNITIVE LAYER — ATTRACTOR CONVERGENCE

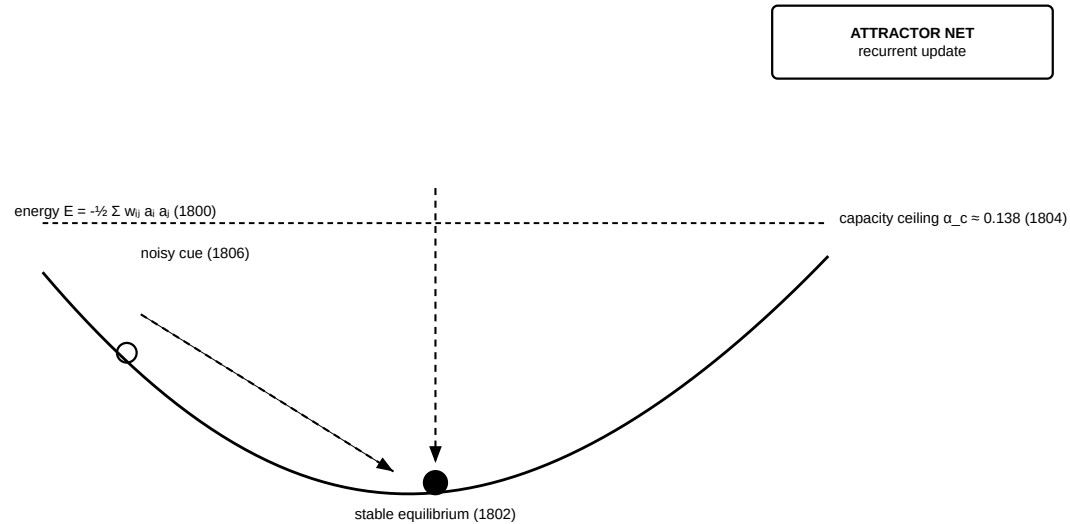
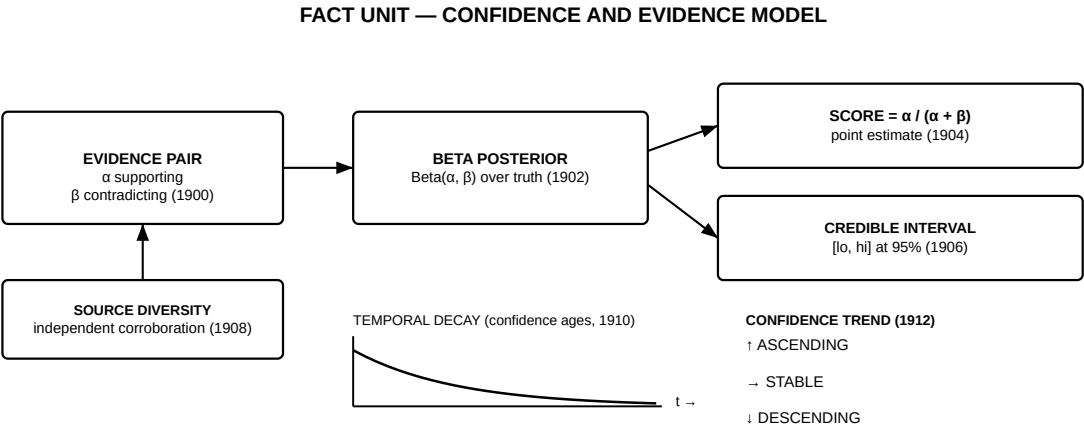


FIG. 26

FIG. 26 — Neuromorphic cognitive layer — attractor convergence

FIG. 27



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 27

FIG. 27 — Fact Unit — confidence and evidence model

FIG. 28

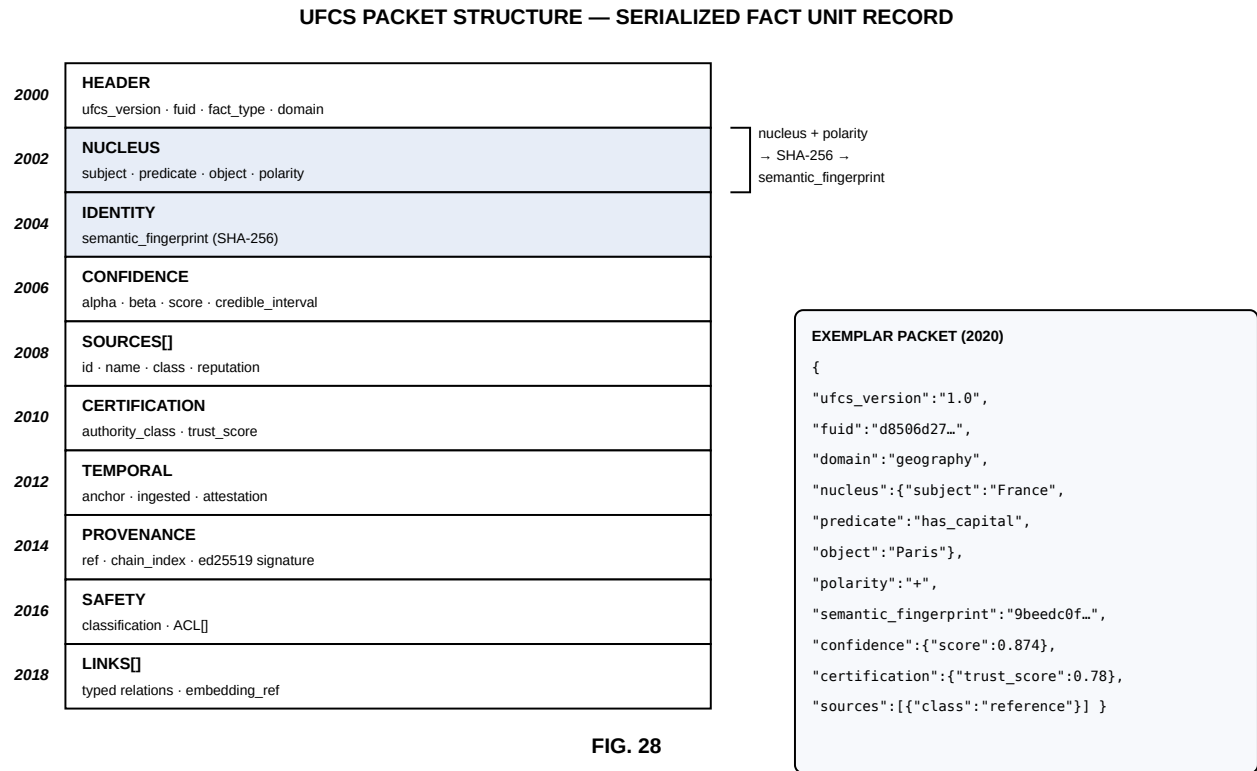


FIG. 28 — UFCS packet structure — serialized Fact Unit record

FIG. 29

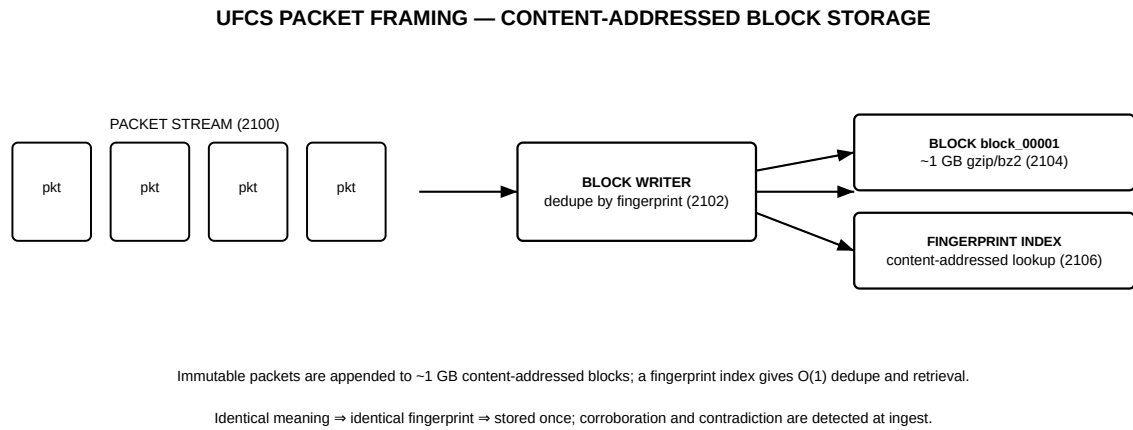


FIG. 29 — UFCS packet framing — content-addressed block storage

FIG. 30

FQL — QUERY ANATOMY AND WORKED EXAMPLE

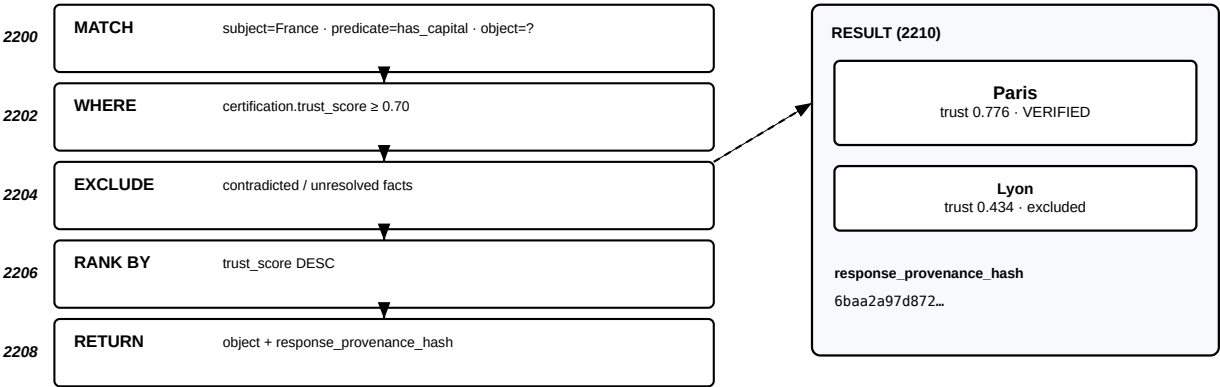


FIG. 30

FIG. 30 — FQL — query anatomy and worked example

FIG. 31

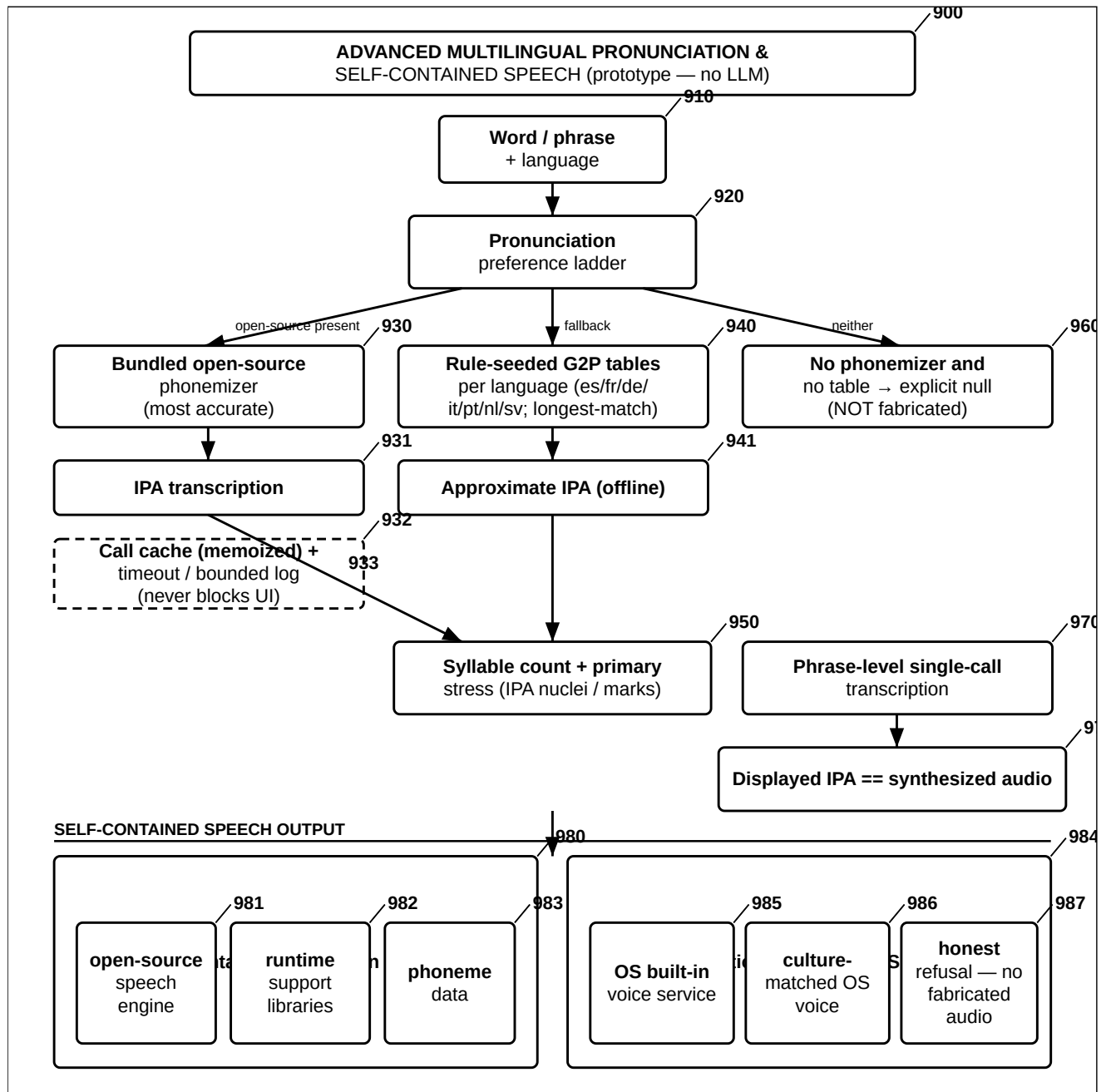


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

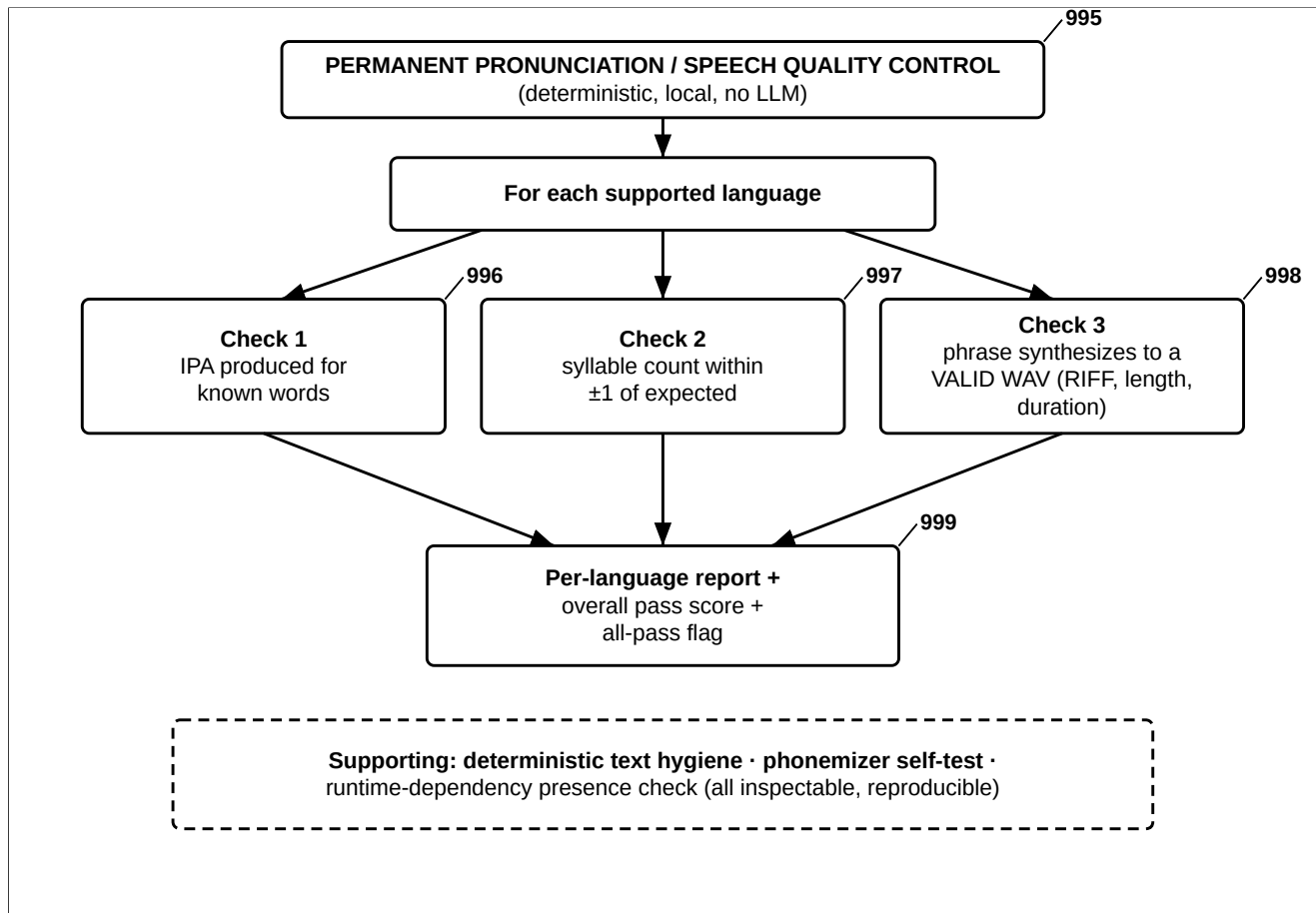


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

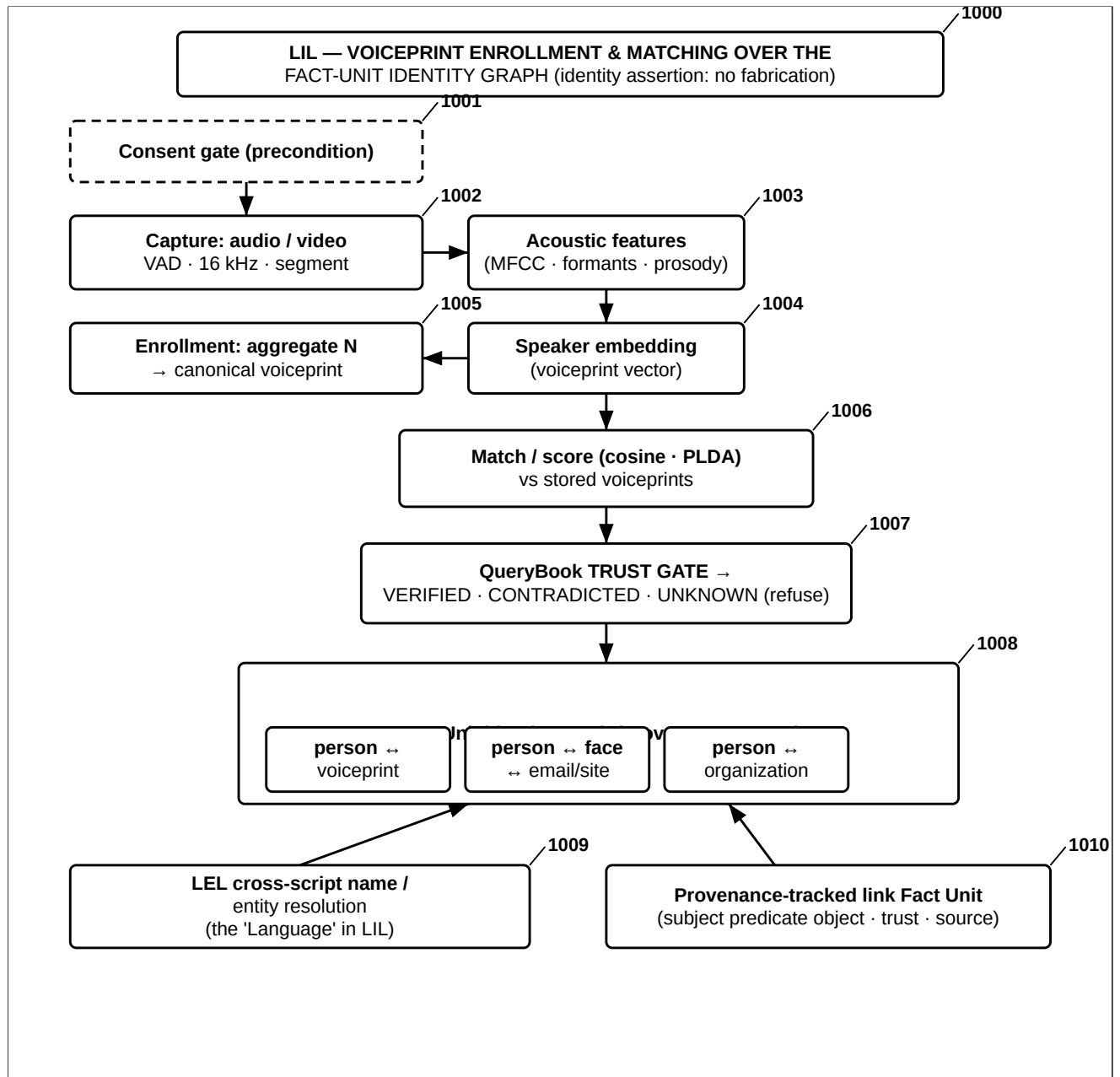


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

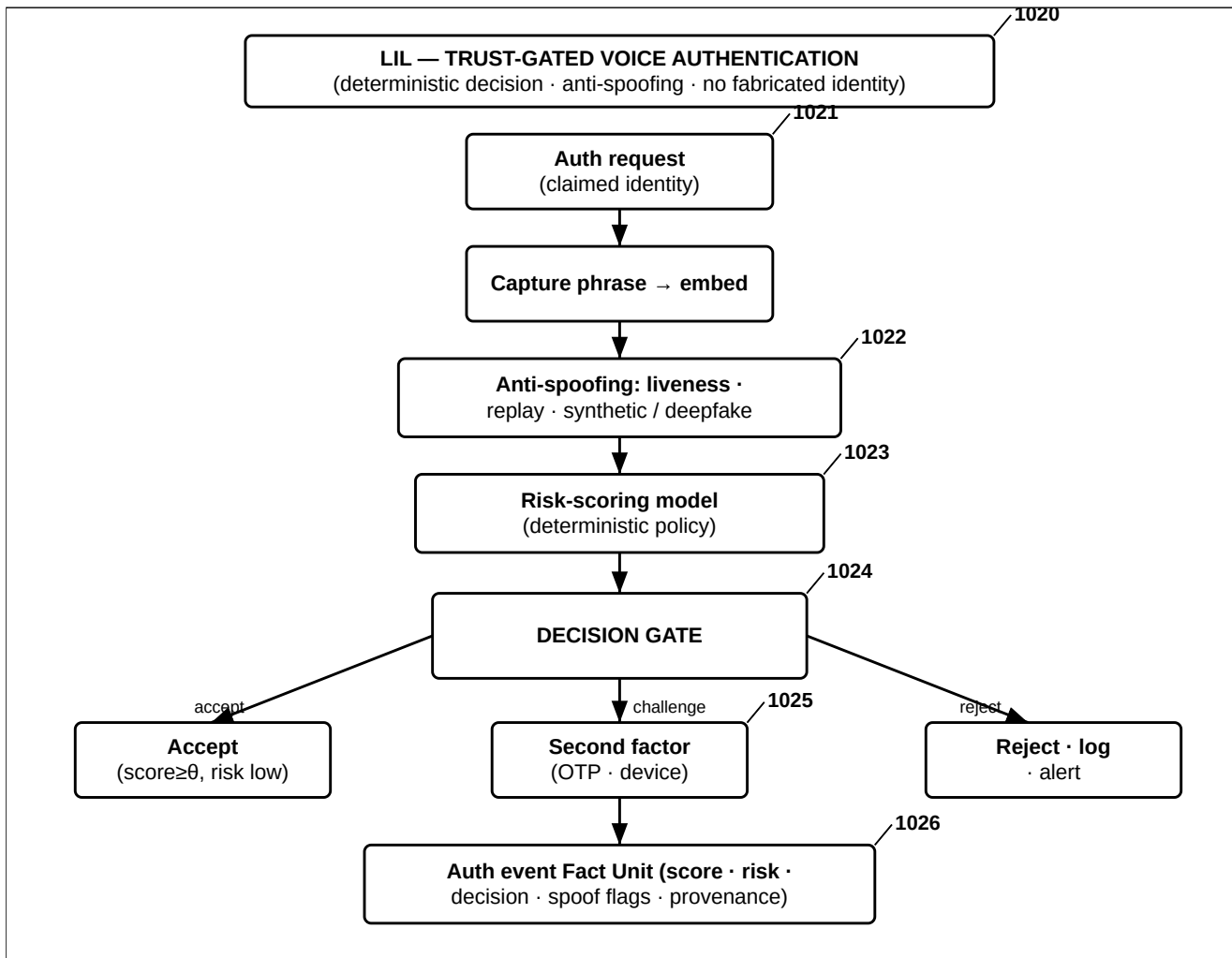


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

FIG. 35

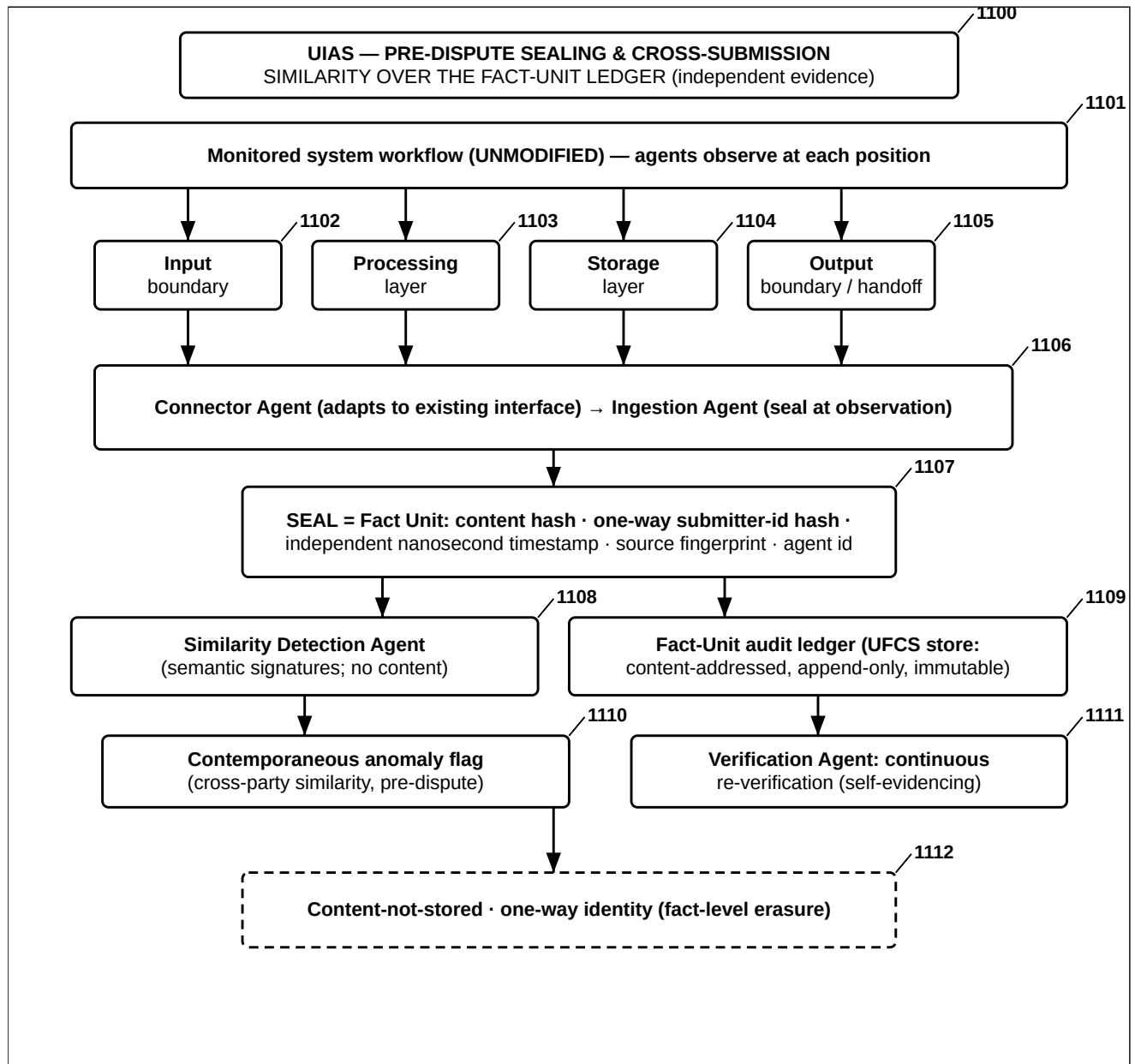


FIG. 35 — UIAS — Pre-Dispute Sealing and Cross-Submission Similarity over the Fact-Unit Ledger

FIG. 36

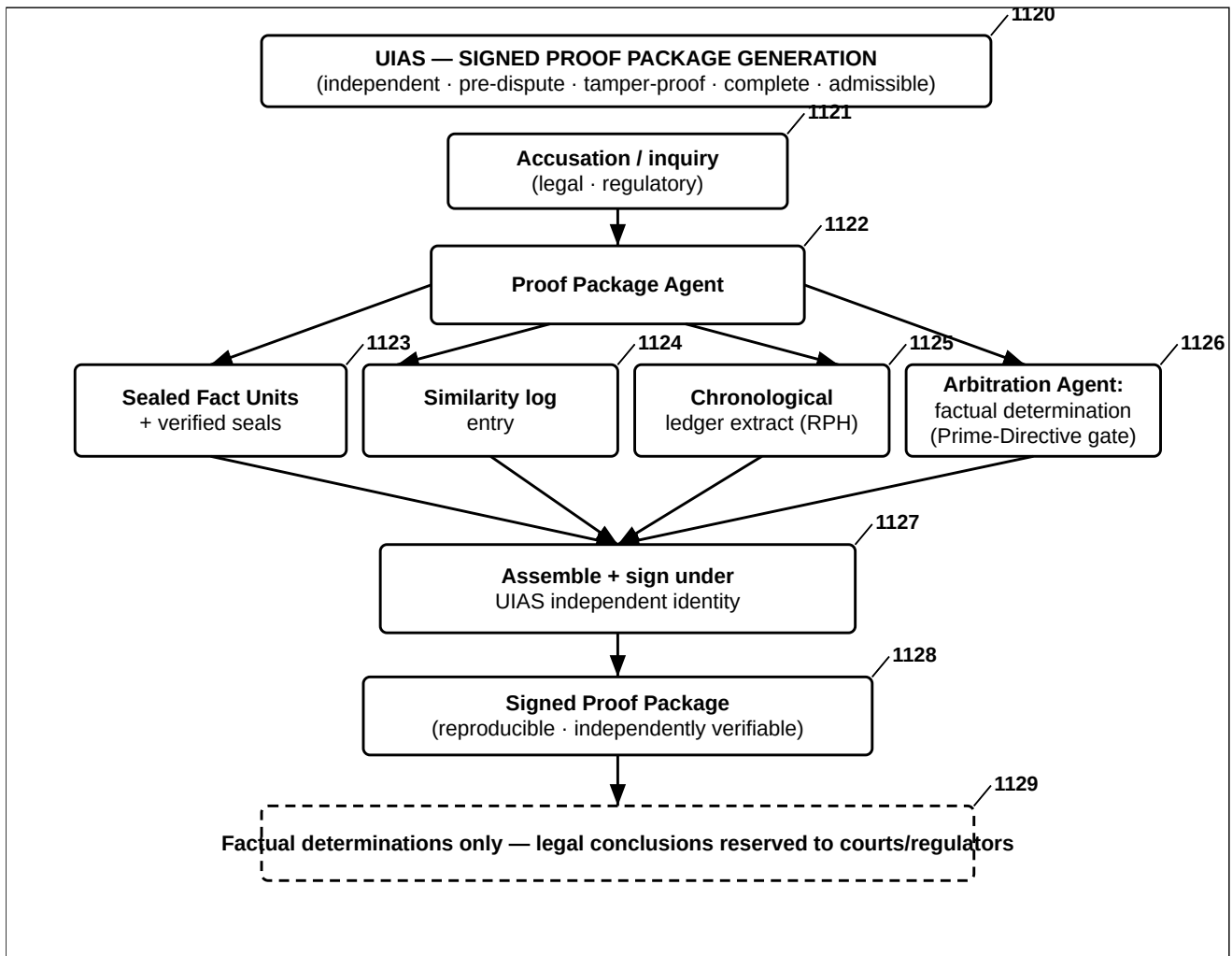


FIG. 36 — UIAS — Signed Proof Package Generation

FIG. 37

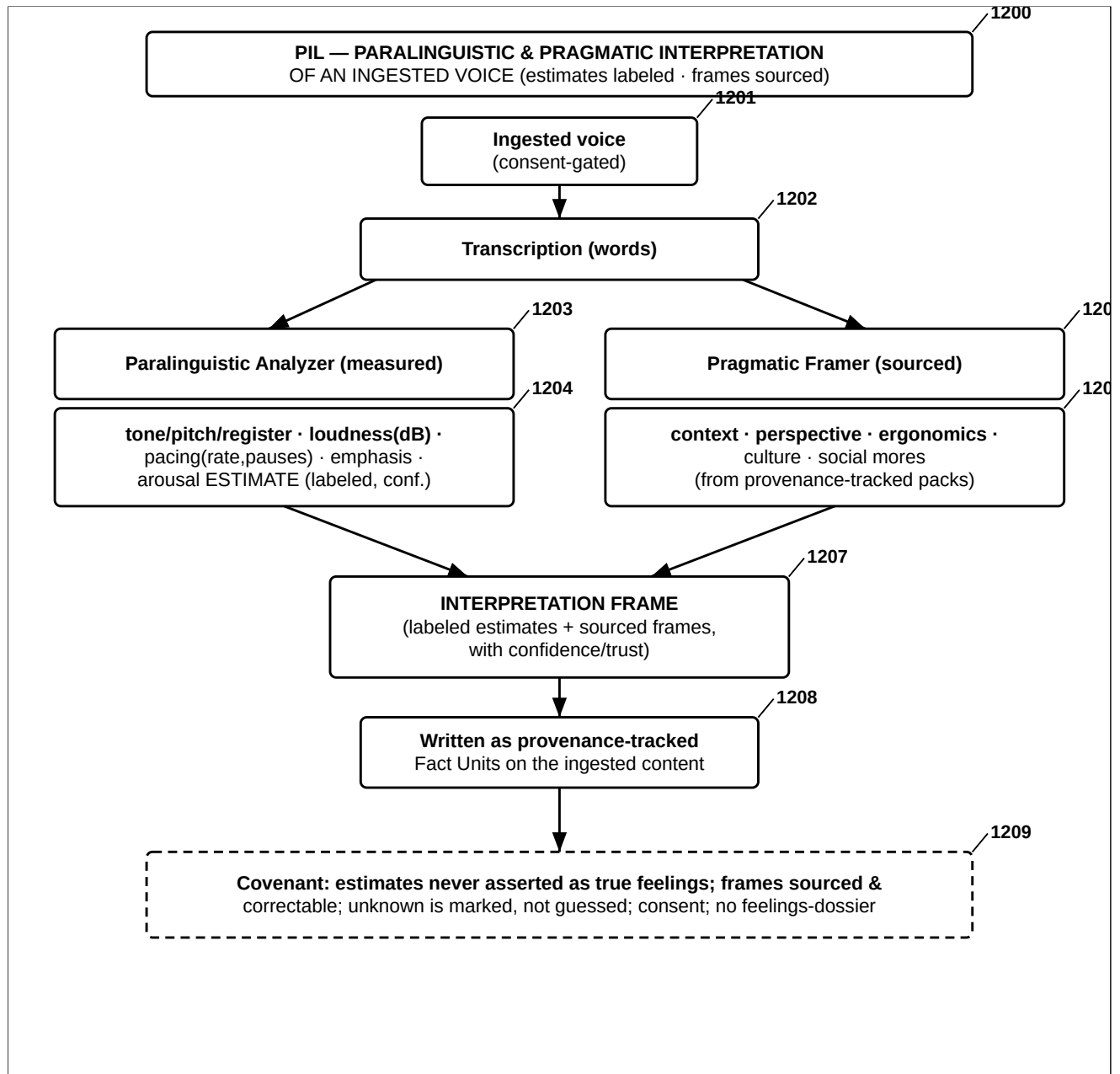


FIG. 37 — PIL — Paralinguistic and Pragmatic Interpretation of an Ingested Voice

FIG. 38

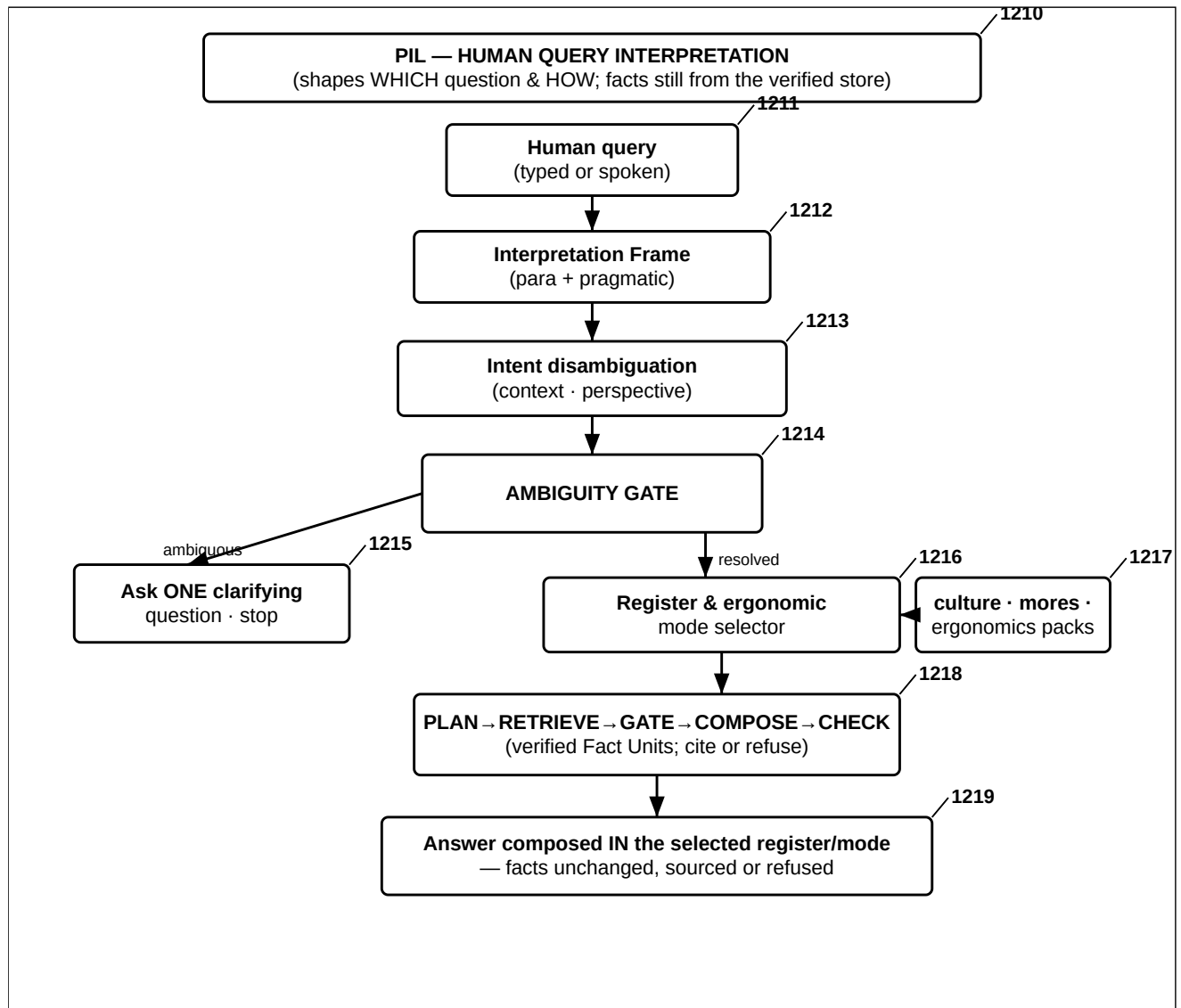


FIG. 38 — PIL — Human Query Interpretation and Covenant-Bound Answering

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Companion drawings volume to the consolidated QueryBook Provisional Patent Application; figures and numerals are drawn from, and identical to, that application. Finalize to 37 CFR 1.84 before filing.